WORKED SOLUTION

STEP II 1987, Question 5

Pure • Lagrange Inversion Series

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q5

The key idea

Lagrange's identity is a machine. You feed it the function $\phi(y) = y - \mathrm{f}(y)$, raise it to the nth power, differentiate $n-1$ times, and add up. All the work is in setting the problem up so that ϕ is simple, ideally a single power of y or a single exponential, because then the $(n-1)$th derivative of ϕ^n can be written down at once. In each part we choose f so that the equation to be solved reads $\mathrm{f}(x) = y$ for a convenient value of y.

Solution

Throughout, write $\phi(y) = y - \mathrm{f}(y)$, so the identity says

$$\mathrm{f}^{-1}(y) = y + \sum_{n=1}^{\infty} \frac{1}{n!} \frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} [\phi(y)^n].$$

Note that the $n = 1$ term involves the zeroth derivative, which is just $\phi(y)$ itself.

The one fact we use repeatedly is that, for a whole number $m \geq n-1$,

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} y^m = \frac{m!}{(m-n+1)!} y^{m-n+1},$$

since each differentiation brings down the current power and lowers it by one.

Part (i)

With $\mathrm{f}(x) = \alpha x$ we have $\phi(y) = (1-\alpha)y$, so $\phi(y)^n = (1-\alpha)^n y^n$. Differentiating $n-1$ times, with $m = n$ above,

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}} [(1-\alpha)^n y^n] = (1-\alpha)^n \, n! \, y.$$

The $n!$ cancels the $\frac{1}{n!}$, so the right-hand side of the identity is

$$y + \sum_{n=1}^{\infty} (1-\alpha)^n y = y \sum_{n=0}^{\infty} (1-\alpha)^n.$$

This is a geometric series with common ratio $1-\alpha$. Since $0<\alpha<2$ we have $-1<1-\alpha<1$, so it converges, and its sum is

$$y\cdot\frac{1}{1-(1-\alpha)}=\frac{y}{\alpha}.$$

On the other hand, $y=\alpha x$ gives $x=y/\alpha$ directly, so $\mathrm{f}^{-1}(y)=y/\alpha$. The two agree, which verifies the identity in this case. The condition $0<\alpha<2$ is exactly the condition for the series to converge.

Part (ii)

The equation is $\mathrm{f}(x)=\frac{1}{2}$ with

$$\mathrm{f}(x)=x-\tfrac{1}{4}x^3,\qquad\text{so}\qquad\phi(y)=y-\mathrm{f}(y)=\tfrac{1}{4}y^3.$$

Then $\phi(y)^n=4^{-n}y^{3n}$, and using the derivative formula with $m=3n$ (which is at least $n-1$),

$$\frac{1}{n!}\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}}[4^{-n}y^{3n}]=\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

So Lagrange's identity gives

$$\mathrm{f}^{-1}(y)=y+\sum_{n=1}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

The leading y is exactly what this general term gives when $n=0$, since $\frac{0!}{0!\,1!\,4^0}\,y^1=y$. So we can absorb it into the sum and start from $n=0$:

$$\mathrm{f}^{-1}(y)=\sum_{n=0}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,4^n}\,y^{2n+1}.$$

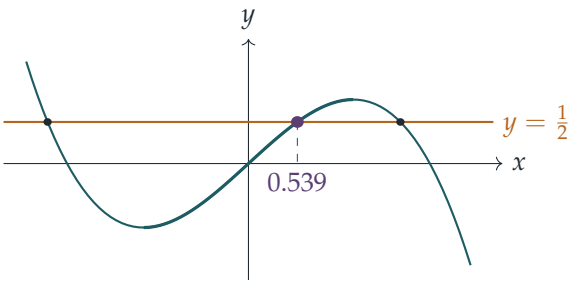
Now put $y=\frac{1}{2}$. Then $4^n\cdot 2^{2n+1}=2^{2n}\cdot 2^{2n+1}=2^{4n+1}$, so

$$x=\mathrm{f}^{-1}(\tfrac{1}{2})=\sum_{n=0}^{\infty}\frac{(3n)!}{n!\,(2n+1)!\,2^{4n+1}},$$

and since $\mathrm{f}(x)=\frac{1}{2}$ this x is a root of $\frac{1}{2}=x-\frac{1}{4}x^3$, as required.

Ask yourself The cubic has three real roots. Which one is the series?

The function f isn't one-to-one on the whole real line, so "f^{-1}" means the branch the series describes. The series is a power series in y that starts $y+\cdots$, so it gives $\mathrm{f}^{-1}(0)=0$ and $\mathrm{f}^{-1}(y)\approx y$ for small y. It is the inverse of f restricted to the increasing piece through the origin, where $\mathrm{f}'(x)=1-\frac{3}{4}x^2>0$, that is, $|x|<2/\sqrt{3}$. The first few terms are $\frac{1}{2}+\frac{1}{32}+\frac{3}{512}+\cdots\approx 0.537$, and the sum is 0.5392 to four decimal places: the smallest positive root, marked in the sketch below. The other roots, about 1.675 and -2.214, lie on the decreasing parts of the curve.



The thicker stretch of the curve is the increasing branch through the origin; the series gives the root on it, shown in plum.

Part (iii)

Here the trick is to put the whole equation into the form $\mathrm{f}(x) = y$ with something simple for ϕ. Rearranging $x = \mathrm{e}^{\lambda x}$ as

$$x - \mathrm{e}^{\lambda x} = 0,$$

we take

$$\mathrm{f}(x) = x - \mathrm{e}^{\lambda x}, \qquad \phi(y) = y - \mathrm{f}(y) = \mathrm{e}^{\lambda y},$$

and we want $x = \mathrm{f}^{-1}(0)$. Now $\phi(y)^n = \mathrm{e}^{n\lambda y}$, and each differentiation multiplies this by $n\lambda$, so

$$\frac{\mathrm{d}^{n-1}}{\mathrm{d}y^{n-1}}\,\mathrm{e}^{n\lambda y} = (n\lambda)^{n-1}\mathrm{e}^{n\lambda y}.$$

(For $n = 1$ this reads $\mathrm{e}^{\lambda y}$, correctly.) Lagrange's identity gives

$$\mathrm{f}^{-1}(y) = y + \sum_{n=1}^{\infty} \frac{(n\lambda)^{n-1}}{n!}\,\mathrm{e}^{n\lambda y},$$

and putting $y = 0$,

$$x = \sum_{n=1}^{\infty} \frac{n^{n-1}}{n!}\,\lambda^{n-1} = 1 + \lambda + \frac{3}{2}\lambda^2 + \frac{8}{3}\lambda^3 + \frac{125}{24}\lambda^4 + \cdots.$$

Equivalently, shifting the index, $x = \displaystyle\sum_{n=0}^{\infty} \frac{(n+1)^{n-1}}{n!}\lambda^n$.

For which λ does this converge? The ratio of consecutive terms is

$$\frac{(n+1)^n\lambda^n/(n+1)!}{n^{n-1}\lambda^{n-1}/n!} = \lambda\left(1 + \frac{1}{n}\right)^{n-1} \to \mathrm{e}\lambda,$$

so by the ratio test the series converges for $|\lambda| < 1/\mathrm{e}$. This is a natural boundary: $y = x$ is tangent to $y = \mathrm{e}^{\lambda x}$ exactly when $\lambda = 1/\mathrm{e}$ (at $x = \mathrm{e}$), and for $\lambda > 1/\mathrm{e}$ there is no real root at all. For $0 < \lambda < 1/\mathrm{e}$ there are two positive roots, and the series gives the smaller one, the one that tends to 1 as $\lambda \to 0$.

Checks

- In (ii), the first four terms are $\frac{1}{2}$, $\frac{1}{32}$, $\frac{3}{512}$ and $\frac{3}{2048}$. Summing 200 terms numerically gives 0.539189, and Newton's method on $x^3 - 4x + 2 = 0$ gives the root $0.5391888\ldots$, in agreement to machine precision.
- In (iii), put $\lambda = 0$: the series gives $x = 1$, and indeed $x = \mathrm{e}^0 = 1$. Summing the series for $\lambda = 0.2$ gives $x \approx 1.29586$, and $\mathrm{e}^{0.2 \times 1.29586} = 1.29586$. For $\lambda = -0.3$ it gives 0.78918, which also satisfies the equation. For $\lambda = 0.2$ the other root is about 12.71, far from the series value.
- In (i), try $\alpha = 1$: then $\phi = 0$, every term of the sum vanishes, and $\mathrm{f}^{-1}(y) = y$, as it should.

Common slips

- Forgetting that the $n = 1$ term has no differentiation (the zeroth derivative), and so dropping or mangling it.
- In (ii), not noticing that the standalone y is the $n = 0$ term of the general formula, and so failing to match the sum, which starts from $n = 0$.
- In (iii), trying $\mathrm{f}(x) = x\mathrm{e}^{-\lambda x}$ or similar. That is legitimate, but then ϕ is a difference of two terms and the derivatives are a mess. Choosing $\phi = \mathrm{e}^{\lambda y}$ makes them trivial.

Beyond STEP

This is the *Lagrange inversion theorem*, which is usually proved with complex analysis (Cauchy's residue theorem) or with formal power series. The coefficients $(n+1)^{n-1}/n!$ in (iii) belong to the Lambert W function, and n^{n-1} counts rooted labelled trees on n vertices. Similarly, the numbers $\frac{(3n)!}{n!\,(2n+1)!}$ in (ii) count ternary trees, a cousin of the Catalan numbers. So Lagrange inversion is a central tool of enumerative combinatorics. Needham's *Visual Complex Analysis* gives the complex-analytic background, and Siklos's *Advanced Problems in Mathematics* has more STEP practice with series.

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